

Blinkit Sales Analysis Project Report

Project Overview

End-to-end data analytics project using Python, PostgreSQL and Power BI.

Objectives

Clean raw data, perform SQL analysis, build an interactive dashboard and derive business insights.

Python Data Cleaning

Handled missing values, standardized column names, feature engineering, data validation.

SQL Analysis

Solved business questions related to revenue, outlet performance, ratings, visibility and product categories.

Power BI Dashboard

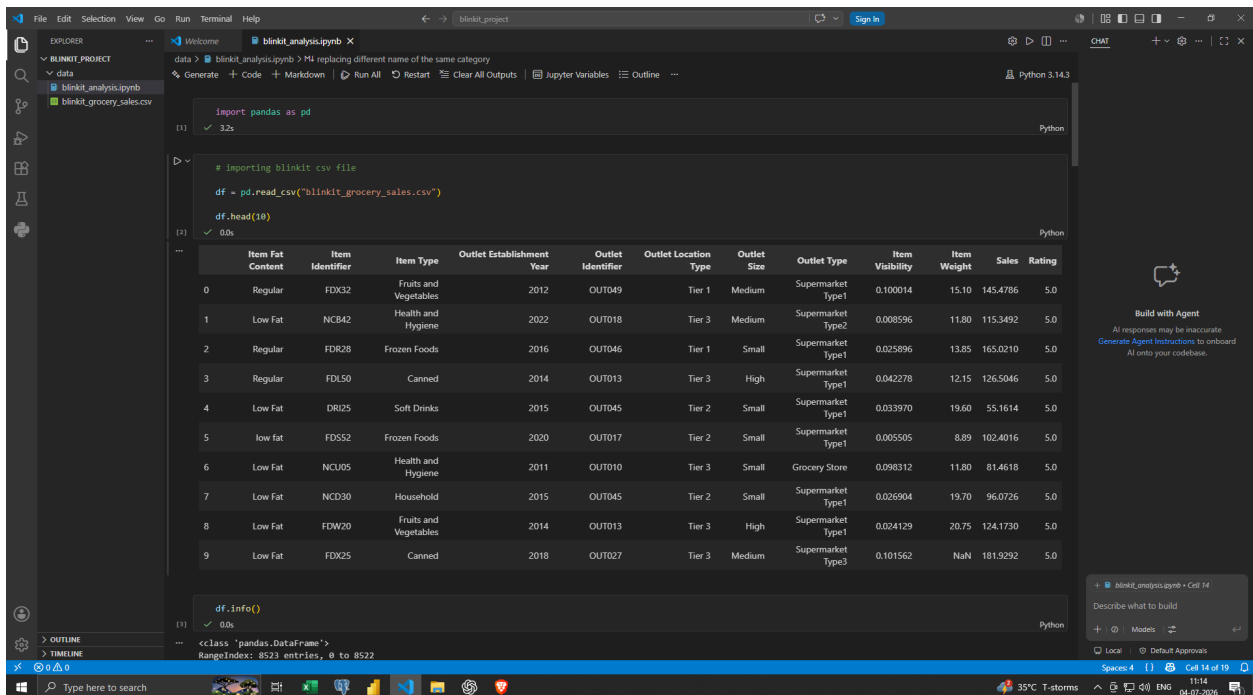
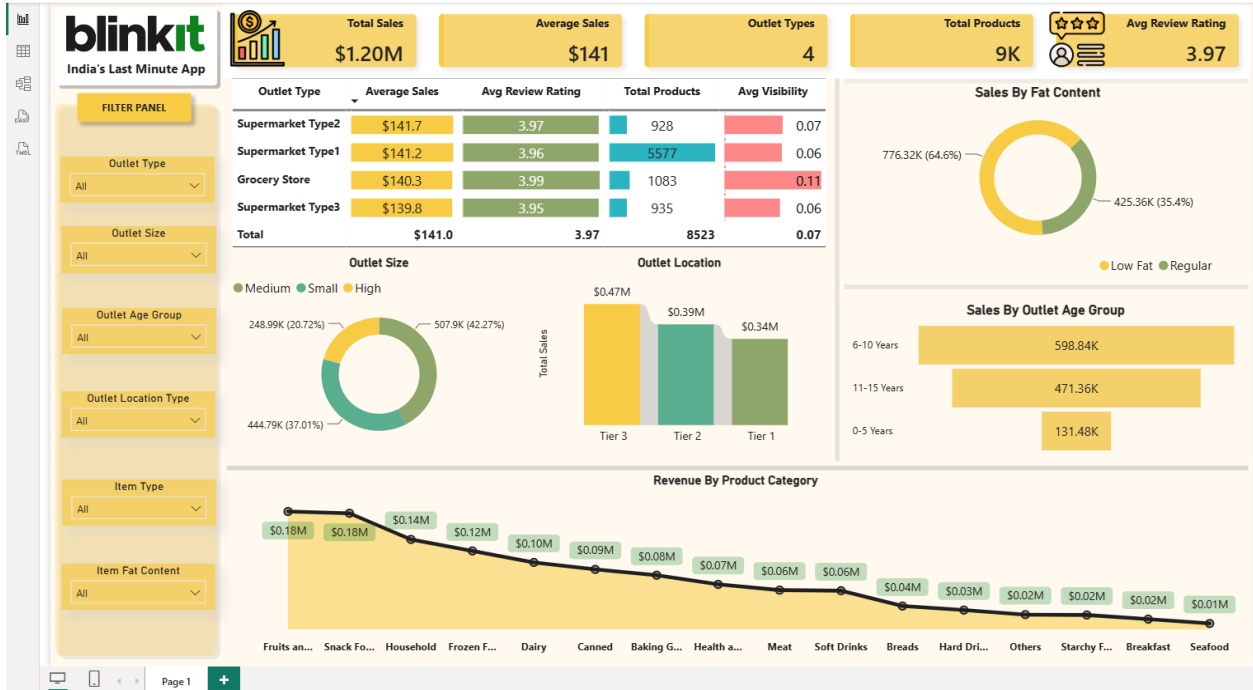
Created KPI cards, slicers, matrix, funnel chart, donut charts and category analysis.

Key Insights

- Tier 3 outlets generated the highest sales.
- Supermarket Type 1 dominated product count.
- Medium outlet sizes performed strongly.
- Low Fat products contributed significantly.
- Outlets aged 6–10 years generated the highest sales.

Tools

Python, Pandas, PostgreSQL, SQL, Power BI, GitHub



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blinkit_project

blinkit_analysis.ipynb

data > blinkit_analysis.ipynb > H4 replacing different name of the same category

Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline Python 3.14.3

item_weight 1463

sales 0

rating 0

dtype: int64

The 'item_weight' column contained missing values that could impact downstream analysis. Since product weight is a fixed attribute for each product, missing values were first imputed using the average weight of the corresponding 'item_identifier'. Any remaining missing values were then filled using the median weight of the respective 'item_type' to maintain realistic product characteristics while minimizing bias.

markdown

df['item_weight'] = df.groupby('item_identifier')['item_weight'].transform(lambda x: x.fillna(x.mean()))

df.isnull().sum()

03

item_fat_content 0

item_identifier 0

item_type 0

outlet_establishment_year 0

outlet_identifier 0

outlet_location_type 0

outlet_size 0

outlet_type 0

item_visibility 0

item_weight 4

sales 0

rating 0

dtype: int64

remaining null value in 'item_weight' is filled by the median weight of corresponding 'item_type'

df['item_weight'] = df.groupby('item_type')['item_weight'].transform(lambda x: x.fillna(x.median()))

df['item_weight'].isnull().sum()

00

np.int64(0)

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

blinkit_analysis.ipynb - Cell 14

Describe what to build

Local Default Approvals

Spaces: 4

Call 14 of 19

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04-07-2026

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blinkit_project

blinkit_analysis.ipynb

data > blinkit_analysis.ipynb > H4 checking unique values in 'item_fat_content' columns to identify inconsistent name conventions and spelling variations. > df['item_fat_content'].unique()

Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline Python 3.14.3

5 low fat FDS52 Frozen Foods 2020 OUT017 Tier 2 Small Supermarket Type1 0.005505 8.89 102.4016 5.0

6 Low Fat NCU05 Health and Hygiene 2011 OUT010 Tier 3 Small Grocery Store 0.098312 11.80 81.4618 5.0

7 Low Fat NCD30 Household 2015 OUT045 Tier 2 Small Supermarket Type1 0.026904 19.70 96.0726 5.0

8 Low Fat FDW20 Fruits and Vegetables 2014 OUT013 Tier 3 High Supermarket Type1 0.024129 20.75 124.1730 5.0

9 Low Fat FDZ25 Canned 2018 OUT027 Tier 3 Medium Supermarket Type3 0.101562 16.70 181.9292 5.0

checking unique values in 'item_fat_content' columns to identify inconsistent name conventions and spelling variations.

markdown

df['item_fat_content'].unique()

00

<StringArray>

['Regular', 'Low Fat', 'low fat', 'LF', 'reg']

length: 5, dtype: str

replacing different name of the same category

markdown

df['item_fat_content'] = df['item_fat_content'].replace({'LF': 'Low Fat', 'low fat': 'Low Fat', 'reg': 'Regular'})

checking again

print(df['item_fat_content'].unique())

00

<StringArray>

['Regular', 'Low Fat']

length: 2, dtype: str

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

blinkit_analysis.ipynb - Cell 13

Describe what to build

Local Default Approvals

Spaces: 4

Call 13 of 19

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blinkit_project

blinkit_analysis.ipynb

data > blinkit_analysis.ipynb > Hi checking unique values in 'item_fat_content' columns to identify inconsistent name conventions and spelling variations. > df["item_fat_content"].unique()

```
print(df["item_fat_content"].unique())
```

0.0s

<StringArray
['Regular', 'Low Fat']
Length: 2, dtype: str

Investigating products with zero visibility values.

df[df["item_visibility"] == 0][["item_identifier", "item_type", "sales", "item_visibility"].head()

0.0s

item_identifier	item_type	sales	item_visibility
20	F0207 Fruits and Vegetables	60.2194	0.0
59	F0C44 Fruits and Vegetables	111.9518	0.0
79	F0V26 Dairy	194.2794	0.0
98	N0K29 Health and Hygiene	123.4730	0.0
113	F0E41 Frozen Foods	83.7566	0.0

products with zero visibility were found to have recorded sales,
suggesting that zero values likely represent missing or incorrectly recorded data.
these values were replaced with median visibility of products within the same item category

```
df["item_visibility"] = df.groupby("item_type")["item_visibility"].transform(  
    lambda x: x.replace(0, x > 0).median())
```

0.0s

Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard
AI onto your codebase.

blinkit_analysis.ipynb - Cell 13
Describe what to build
+ Modes
Local Default Approvals
Spaces: 4 | Call 13 of 19
11:16
04-07-2026

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blinkit_project

blinkit_analysis.ipynb

data > blinkit_analysis.ipynb > Hi creating outlet_age to analyze whether older outlets perform differently from newer outlets. > current_year = 2026

creating sales category based on sales distribution for easier segmentation and dashboard filtering

```
df["sales_category"] = pd.qcut(df["sales"], q=3, labels=["Low", "Medium", "High"])
```

checking

```
df.head()
```

0.0s

item_fat_content	item_identifier	item_type	outlet_establishment_year	outlet_identifier	outlet_location_type	outlet_size	outlet_type	item_visibility	item_weight	sales	rating
Regular	FDX32	Fruits and Vegetables	2012	OUT049	Tier 1	Medium	Supermarket Type1	0.100014	15.10	145.4786	5.0
Low Fat	NCB42	Health and Hygiene	2022	OUT018	Tier 3	Medium	Supermarket Type2	0.008596	11.80	115.3492	5.0
Regular	FDR28	Frozen Foods	2016	OUT046	Tier 1	Small	Supermarket Type1	0.025896	13.85	165.0210	5.0
Regular	FDL50	Canned	2014	OUT013	Tier 3	High	Supermarket Type1	0.042278	12.15	126.5046	5.0
Low Fat	DRJ25	Soft Drinks	2015	OUT045	Tier 2	Small	Supermarket Type1	0.033970	19.60	55.1614	5.0

creating outlet_age to analyze whether older outlets perform differently from newer outlets.

```
current_year = 2026  
df["outlet_age"] = current_year - df["outlet_establishment_year"]
```

#checking

```
df.head()
```

0.0s

item_identifier	item_type	outlet_establishment_year	outlet_identifier	outlet_location_type	outlet_size	outlet_type	item_visibility	item_weight	sales	rating	sales_category	outlet_age
Fruits and						Supermarket						

Build with Agent
AI responses may be inaccurate
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AI onto your codebase.

blinkit_analysis.ipynb - Cell 23
Describe what to build
+ Modes
Local Default Approvals
Spaces: 4 | Call 23 of 23
11:39
04-07-2026

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dashboard_blink1 • Last saved: Today at 1:17 PM

FileHomeHelpTable toolsColumn tools

NameColumnFormat

TypeWhole number%

SummarizationSum

Data categoryUncategorized

Sort by column

Data groups

Manage relationships

New columns

Calculations

Structure

Formating

Properties

Sort

Groups

Relationships

Calculations

1 Outlet Age Group +

2 SWITCH

3 IF(C1,

4 'public blinkit_sales'[outlet_age] <= 5, "0-5 Years",

5 'public blinkit_sales'[outlet_age] <= 10, "6-10 Years",

6 'public blinkit_sales'[outlet_age] <= 15, "11-15 Years",

7 "16+ Years",

8)

Item_fat_contentItem_IdentifierItem_typeoutlet_establishment_yearoutlet_Identifieroutlet_location_typeoutlet_sizeoutlet_TypeItem_visibilityItem_weightsalesratingSales_categoryoutlet_ageColumn

Low fatFDV49Canned2015OUT045Tier 2SmallSupermarket Type10.02587957710265.2264High??

RegularFDV27Meat2015OUT045Tier 2SmallSupermarket Type10.0400711317.9787.35144Low??

RegularFDI03Dairy2015OUT045Tier 2SmallSupermarket Type10.07894645313.65186.0244High??

RegularFDV11Breads2015OUT045Tier 2SmallSupermarket Type10.0586472459.1173.20544High??

Low fatFDI04Baking Goods2015OUT045Tier 2SmallSupermarket Type10.17657303511.1157.46044Medium??

Hard DrinkDRN47Hard Drinks2015OUT045Tier 2SmallSupermarket Type10.04342535112.1179.8664High??

Low fatNCG55Health and Hygiene2015OUT045Tier 2SmallSupermarket Type10.02100261611.5130.79424Medium??

Low fatNCG41Health and Hygiene2015OUT045Tier 2SmallSupermarket Type10.0561038415.7153.50244Medium??

Low fatNCF43Household2015OUT045Tier 2SmallSupermarket Type10.0520512568.51142.2474Medium??

Low fatNCK18Household2015OUT045Tier 2SmallSupermarket Type10.0490165179.6164.21844Medium??

Low fatNCK18Household2015OUT045Tier 2SmallSupermarket Type10.0088181814.15196.5114High??

Low fatNCL06Household2015OUT045Tier 2SmallSupermarket Type10.07201278714.65265.90944High??

Low fatNCE30Household2015OUT045Tier 2SmallSupermarket Type10.04905161712.6210.49024High??

Low fatFDI39Meat2015OUT045Tier 2SmallSupermarket Type10.03611103818.8558.15624Low??

Low fatNCG55Others2015OUT045Tier 2SmallSupermarket Type10.05967115314.6238.35384High??

Low fatFDH09Seafood2015OUT045Tier 2SmallSupermarket Type10.05619130112.653.29824Low??

Low fatFDR33Snack Foods2015OUT045Tier 2SmallSupermarket Type10.0268432657.31109.7574Medium??

Low fatFDI04Snack Foods2015OUT045Tier 2SmallSupermarket Type10.0159084857.945161.4214Medium??

Low fatFDI10Snack Foods2015OUT045Tier 2SmallSupermarket Type10.0295722988.395109.95424Low??

Low fatFDA57Snack Foods2015OUT045Tier 2SmallSupermarket Type10.03972399918.8541.0484Low??

Low fatFDA45Snack Foods2015OUT045Tier 2SmallSupermarket Type10.15569479421.25177.67374High??

Low fatDRF25Soft Drinks2015OUT045Tier 2SmallSupermarket Type10.039004193937.0194Low??

Low fatDRL04Soft Drinks2015OUT045Tier 2SmallSupermarket Type10.11255905811.8184.29244High??

Low fatDRN36Soft Drinks2015OUT045Tier 2SmallSupermarket Type10.05027960415.294.17524Medium??

Low fatDRH37Soft Drinks2015OUT045Tier 2SmallSupermarket Type10.04170075617.6164.85264Medium??

Low fatFDI28Frozen Foods2015OUT045Tier 2SmallSupermarket Type10.06369080413.3151.07084Medium??

Low fatNCP90Household2015OUT045Tier 2SmallSupermarket Type10.03385714720.540.28324Low??

Low fatFDI91Dairy2015OUT045Tier 2SmallSupermarket Type10.0355320626.9265.95524Medium??

RegularFDE04Frozen Foods2015OUT045Tier 2SmallSupermarket Type10.01805962119.75179.7664High??

Low fatFDN57Snack Foods2015OUT045Tier 2SmallSupermarket Type10.05434418618.25140.21544Medium??

Low fatFDL56Fruits and Vegetables2015OUT045Tier 2SmallSupermarket Type10.12603569414.186.41984Low??

Low fatNCR38Household2015OUT045Tier 2SmallSupermarket Type10.1374486517.25251.37244High??

Table: public.blinkit_sales (8,523 rows) Column: Column (1 distinct values)

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FIFA World Cup 2026...

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